Calculus 1 sec 2: Assignment 5, due Thursday April 30, 2026, 11:59pm

Submit the completed assignment to https://www.gradescope.com.
Only hand-written solutions are accepted.

- This assignment is worth a total of 15 points, and accounts for about 3% of your final grade. The standard submission deadline is Thursday April 30, 11:59pm (with an extra no penalty day.) Submissions after the late due date will NOT be accepted unless you have a good excuse.
- Collaboration is allowed, but you must write up your solutions independently and in your own words. Copying from others, publications, or websites (even with minor changes) will be treated as plagiarism.
- Neatness, clarity, and correct notation matter! Show all steps, use clear explanations, and write in complete sentences where needed. Graders will only evaluate what you write, so unclear or incomplete work will affect your grade, even with the correct answer.
- Attempt all questions (and parts), and clearly indicate which question you are answering. Answer each question on a separate sheet, and please **match your answers with the correct questions on Gradescope**.

1 [3 pt] Let R be a rectangle with corners at the points $(\pm a, \pm b)$ where the point (a, b) (with $a, b \geq 0$) lies on the parabola $y = 1 - x^2$. What is its maximum area? What shape does it have at the maximum? Draw a diagram.

2. [3 pt] A canister is dropped from a helicopter that is 400m. above ground, and falls vertically. Unfortunately its parachute does not open. It can withstand an impact velocity of 90m/sec. Does it survive the impact? (Approximate the acceleration due to gravity by 9.8 m/sec^2.)

3. (i) [.5 pt] Give formulas for two different continuous functions $f : \mathbb{R} \to \mathbb{R}$ that are not identically zero and have the property that $\int_{-x}^{x} f(t)dt = 0$ for all x.
(ii) [.5 pt] Describe the most general continuous function $f : \mathbb{R} \to \mathbb{R}$ with the property that $\int_{-x}^{x} f(t)dt = 0$ for all x.
(iii) [1 pt] Give the formula of a nonconstant continuous function such that $\int_{-x}^{2x} f(t)dt = 0$ for all x.

4 (a) [2 pt] Find a function f and a number $a \in (0, \pi)$ such that $\frac{\pi}{2} + \int_{a}^{x} f(t)dt = x \sin(x)$ for all x. **Hint:** Differentiate this equation with respect to x.

(b) [2 pt] Find all the points x where the function $f(x) = \int_{0}^{x^2} \frac{t-3}{2-\sin(3t)} dt$ has a local max/min, and say which are max., which are min.

5 (a) [1 pt] Find the slope of the line perpendicular to the curve $xy = 1$ at the point $(a, 1/a)$.
(b) [2 pt] Show that if P is the point on the curve $xy = 1$ that is closest to the point $Q = (0, 2)$, then the line QP is perpendicular to the curve $xy = 1$. (You do not need to find the coordinates of P.)